

FloodSentinel: Flood Risk Assessment Using Multi Temporal Satellite Imagery and Deep Neural Networks

LITERATURE REVIEW

1. A Fares Hamad Aljohani, Ahmad. B. Alkhodre, Adnan Ahamad Abi Sen, Muhammad Sher Ramazan, Bandar Alzahrani and Muhammad Shoaib Siddiqui, "Flood Prediction using Hydrologic and ML-based Modeling: A Systematic Review" International Journal of Advanced Computer Science and Applications(IJACSA), 14(11), 2023, <http://dx.doi.org/10.14569/IJACSA.2023.0141155>.

Existing approaches leverage *hydrologic models* (1D/2D/3D) to simulate physical processes like rainfall-runoff dynamics and terrain interactions, and *machine learning (ML) techniques* (e.g., ANNs, SVMs, WNNs) to detect complex patterns in historical flood data for predictive accuracy. **However, they fall short because** hydrologic models require extensive geospatial data, expert calibration, and significant computational resources, limiting real-time application and scalability, while ML models lack inherent physical interpretability, struggle to extrapolate accurately to extreme or unprecedented events, and often fail to integrate fundamental hydrological constraints. **Therefore, there is a need to address** this limitation in robustness and generalizability **by developing** a *hybrid framework* that strategically integrates physics-based hydrologic modeling with adaptive ML algorithms, using data fusion techniques to leverage their complementary strengths for reliable, interpretable flood prediction in dynamic environments.

2. Dimara Kusuma Hakim, Rahmat Gernowo, Anang Widhi Nirwansyah, Flood prediction with time series data mining: Systematic review, Natural Hazards Research, Volume 4, Issue 2, 2024, Pages 194-220, ISSN 2666-5921, <https://doi.org/10.1016/j.nhres.2023.10.001>.

Existing approaches primarily utilize *univariate time series analysis* to forecast floods based on single indicators (e.g., rainfall or river levels), and *multivariate time series methods* (48.62% of studies) to incorporate multiple environmental factors, demonstrating improved predictive capability for certain flood types like river floods. **However, they fall short because** (1) they overwhelmingly focus on river floods (76.1%) and urban floods (11.9%), neglecting critical but complex flood types like coastal floods (6.4%) and flash floods (5.5%); (2) they underutilize optimization techniques (only 11% of studies) for refining model parameters (e.g., neural network weights or damping factors), limiting accuracy; and (3) they rely heavily on scale-dependent metrics like RMSE, failing to provide intuitive, percentage-based evaluations (e.g., MAPE) for end-user decision-making. **Therefore, there is a need to address** these limitations in coverage, optimization, and usability **by developing** a *hybrid, optimized time-series model* that specifically targets underrepresented flood types (coastal and flash floods), integrates advanced optimization algorithms (e.g., PSO or GRG Nonlinear) for parameter calibration, and employs user-centric evaluation metrics like MAPE to enhance real-world applicability for early warning systems and infrastructure planning.

3. Kadiyala, Sai Prasanth, and Wai Lok Woo. "Flood prediction and analysis on the relevance of features using explainable artificial intelligence." *arXiv preprint arXiv:2201.05046* (2022), <https://arxiv.org/abs/2201.05046>.

Existing approaches leverage *traditional machine learning models* (e.g., Logistic Regression, SVM, Decision Trees) to achieve high accuracy in binary flood prediction and employ *post-hoc explainability techniques* (e.g., SHAP, LIME) to interpret feature contributions. **However, they fall short because** (1) they rely exclusively on shallow ML algorithms, ignoring the potential of deep learning for complex pattern recognition in flood drivers; (2) their explanations remain superficial, failing to translate feature relevance into actionable mitigation strategies; and (3) they treat explainability as a separate validation

step rather than an integrated component for model refinement. **Therefore, there is a need to address** these limitations in model sophistication and practical utility **by developing** an *end-to-end interpretable framework* that combines deep neural architectures with *customized explainability modules* to not only reveal critical flood indicators (e.g., rainfall thresholds, spatial dependencies) but also generate prescriptive insights for disaster planners, such as infrastructure prioritization or evacuation triggers.

4. Chenmin, Ni, Muhammad Fadhil Marsani, and Fam Pei Shan. "Flood prediction based on feature selection and a hybrid deep learning network." *Journal of Water and Climate Change* 15.3 (2024): 1245-1261, <https://doi.org/10.2166/wcc.2024.559>.

Existing approaches leverage *Variational Mode Decomposition (VMD)* for denoising hydrological data and combine it with *deep learning models* (e.g., LSTM, BiLSTM, CNN) to capture temporal dependencies and spatial features in flood forecasting. **However, they fall short because** (1) VMD parameter selection lacks systematic classification of intrinsic mode functions (IMFs), leading to inconsistent decomposition outcomes; (2) decomposed IMFs exhibit distinct probability distributions and variable correlations, yet traditional methods directly feed all features into monolithic models without dynamic feature matching; and (3) integrated meteorological-hydrological inputs are processed indiscriminately, causing overfitting on stable signals (low-frequency IMFs) and underfitting on volatile components (high-frequency IMFs). **Therefore, there is a need to address** these limitations in adaptability and feature-model alignment **by developing** a *hybrid VMD-F framework* that: Classifies IMFs using a novel *fluctuation factor* to segregate low-frequency (smooth) and high-frequency (volatile) components; Dynamically matches features via *Pearson correlation* to select optimal meteorological inputs (e.g., rainfall, temperature) for each IMF category; Integrates specialized submodels (CNN-BiLSTM-Attention for spatial-temporal high-frequency patterns; BiLSTM for long-term low-frequency trends) to exploit component-specific characteristics.

5. Yehai Tang, Yue Sun, Zhenyue Han, Shan-e-hyder Soomro, Qiang Wu, Boyu Tan, Caihong Hu, flood forecasting based on machine learning pattern recognition and dynamic migration of parameters, *Journal of Hydrology: Regional Studies*, Volume 47, 2023, 101406, ISSN 2214-5818, <https://doi.org/10.1016/j.ejrh.2023.101406>.

Existing approaches optimize hydrological model parameters (e.g., Xin'anjiang) using *historical flood clustering* (via K-means, fuzzy logic) and employ *rainfall-runoff simulations* to improve forecast accuracy. **However, they fall short because** (1) fixed parameters calibrated to past events fail to adapt to changing environments (climate shifts/human activities); (2) classification methods rely heavily on historical flood patterns that may not represent *new* extreme events; and (3) traditional frameworks lack real-time mechanisms to dynamically adjust parameters based on *incoming rainfall characteristics*. **Therefore, there is a need to address** this inflexibility in evolving conditions **by developing** a *probabilistic ensemble framework* that: Dynamically classifies emerging flood types using low-dimensional rainfall indicators (intensity/duration), Matches real-time storms to analogous historical clusters via random forest and Generates adaptive forecasts by weighting Xin'anjiang parameters from similar flood types.

6. Ogbuene, E.B., Eze, C.A., Aloh, O.G., Oroke, A.M., Udegbonam, D.O., Ogbuka, J.C., Achoru, F.E., Ozorme, V.A., Anwara, O., Chukwunonye, I., Nebo, A.N. and Okolo, O.J. (2024) Application of Machine Learning for Flood Prediction and Evaluation in Southern Nigeria, *Atmospheric and Climate Sciences*, 14, 299-316, <https://doi.org/10.4236/acs.2024.143019>.

Existing approaches apply *standard ML classifiers* (e.g., Naive Bayes, Random Forest) to sparse historical flood records and rely on *damage assessment metrics* (e.g., property loss, casualties) for model evaluation. **However, they fall short because** (1) they train on extremely limited datasets (e.g., 59 events over 30 years) with severe data access constraints; (2) they ignore critical geospatial drivers

(e.g., terrain, soil saturation, urbanization patterns); and (3) they prioritize classification accuracy over actionable early warning capabilities. **Therefore, there is a need to address** these limitations in data poverty and contextual relevance **by developing a transfer learning framework** that: Integrates satellite-derived hydrometeorological data (e.g., soil moisture, precipitation anomalies) to compensate for sparse ground records, Combines ML with hydrodynamic modeling to capture localized flood dynamics and Generates interpretable risk maps rather than binary flood/non-flood predictions.

7. Yu, J., Li, Y., Huang, X., & Ye, X. (2025). Data quality and uncertainty issues in flood prediction: a systematic review. *International Journal of Digital Earth*, 18(1), <https://doi.org/10.1080/17538947.2025.2495738>.

Existing approaches leverage *diverse data types* (hydrological, topographical, and rainfall patterns) and employ *machine learning hybrids* (e.g., PSO-optimized TCN, VMD-BiLSTM) to improve prediction accuracy. **However, they fall short because** (1) they prioritize model sophistication over **data infrastructure gaps** in developing regions, where 67% of flood damage studies lack local data; (2) they underutilize **real-time dynamic data** (e.g., IoT sensors, crowdsourced reports) that could mitigate spatial-temporal uncertainties; and (3) they neglect **integrated uncertainty quantification** across the modeling chain, despite hydrology being the dominant uncertainty source. **Therefore, there is a need to address** these limitations in inclusivity and adaptability **by developing a transfer learning framework** that: Deploys low-cost IoT networks for real-time hydrological monitoring in data-scarce regions, Integrates AI-driven data fusion (e.g., GANs for synthetic scenario generation, attention mechanisms for CSD noise reduction) and Embeds Bayesian uncertainty propagation within ensemble models to quantify errors from input to output.

8. Nearing, G., Cohen, D., Dube, V. *et al.* Global prediction of extreme floods in ungauged watersheds. *Nature* **627**, 559–563 (2024), <https://doi.org/10.1038/s41586-024-07145-1>.

Existing approaches rely on *physics-based hydrological models* (e.g., GloFAS) calibrated to long local streamflow records and generate *global flood nowcasts (0-day lead)*. **However, they fall short because** they cannot function in ungauged basins (>97% of global watersheds) due to calibration dependencies, disproportionately leaving vulnerable developing regions without reliable forecasts. **Therefore, there is a need to address** this exclusionary limitation **by developing transferable AI architectures** (e.g., LSTM encoder-decoder) trained on heterogeneous global datasets to deliver actionable 5-day lead forecasts in data-scarce regions, democratizing early warnings via open-access platforms.

9. Byaruhanga, N.; Kibirige, D.; Gokool, S.; Mkhonta, G. Evolution of Flood Prediction and Forecasting Models for Flood Early Warning Systems: A Scoping Review. *Water* **2024**, *16*, 1763. <https://doi.org/10.3390/w16131763>.

Existing approaches leverage *coupled hydrologic-hydraulic models* for physical accuracy and employ *machine learning ensembles* (e.g., ANN hybrids) to enhance predictive capabilities in data-rich regions. **However, they fall short because** they critically depend on high-resolution ground data for calibration—a requirement unmet in >70% of developing regions where gauge networks are sparse or non-existent—and impose prohibitive computational costs that hinder real-time implementation in resource-constrained settings. **Therefore, there is a need to address** these accessibility and scalability barriers **by developing a cloud-native, transfer learning framework** that integrates satellite precipitation products (SPPs) with lightweight LSTM architectures, enabling operational FEWSs in data-scarce basins through: Automated SPP-ground data fusion to compensate for missing gauge records, Pre-trained model repositories for rapid regional adaptation and Edge-computing deployment to reduce cloud dependency.

10. Yu, Q., Liu, C., Li, R. *et al.* Research on a hybrid model for flood probability prediction based on time convolutional network and particle swarm optimization algorithm. *Sci Rep* **15**, 6870 (2025), <https://doi.org/10.1038/s41598-024-80100-2>.

Existing approaches leverage *hybrid ML models* (e.g., PSO-optimized TCN) to enhance prediction accuracy and employ *Bootstrap sampling* for uncertainty quantification. **However, they fall short because** they fail to maintain acceptable error thresholds (<20%) beyond 5-hour lead times due to insufficient integration of physical flood mechanisms, limiting model generalizability under extreme climate variability. **Therefore, there is a need to address** this predictive decay **by developing** a *physics-informed deep learning framework* that: Embeds hydrological constraints (e.g., soil moisture dynamics, basin lag times) directly into TCN architectures, Incorporates climate-driven covariates (urbanization impacts, circulation patterns) as model inputs and Uses multi-objective optimization to balance reliability/concentration in probabilistic outputs.

11. Hajji Sonia , Krimissa Samira , Abdelrahman Kamal , Boudhar Abdelghani , Elaloui Abdenbi , Ismaili Maryam , El Bouzekraoui Meryem , Chikh Essbiti Mohamed , Kahal Ali Y. , Mondal Biraj Kanti , Namous Mustapha, Enhancing flood prediction through remote sensing, machine learning, and Google Earth Engine, *Frontiers in Water*, Volume 7, 2025, 10.3389/frwa.2025.1514047, <https://www.frontiersin.org/journals/water/articles/10.3389/frwa.2025.1514047>.

Existing approaches leverage *spectral indices* (e.g., NDFI) for automated flood inventory mapping and employ *ensemble ML models* (RF, GB) with SHAP explainability to assess flood susceptibility using topographic/hydrologic factors. **However, they fall short because** (1) optical indices like NDFI fail under cloud cover and dense vegetation—critical limitations during flood-triggering storms; (2) susceptibility mapping ignores dynamic flood mechanisms (e.g., flow velocity, infiltration capacity); and (3) models prioritize spatial risk over temporal forecasting, preventing real-time warnings. **Therefore, there is a need to address** these static and sensor limitations **by developing** a *multi-sensor fusion framework* that: Integrates microwave remote sensing (Sentinel-1 SAR) with NDFI to penetrate clouds/vegetation, Embeds hydrodynamic constraints (e.g., soil porosity, rainfall intensity) into ML architectures and Generates time-resolved risk scores using LSTM networks fed by real-time gauge/weather data.

12. Almikael, W., Šoltész, A., Čubanová, L. *et al.* Hydro-informer: a deep learning model for accurate water level and flood predictions. *Nat Hazards* **121**, 3959–3979 (2025). <https://doi.org/10.1007/s11069-024-06949-8>.

Existing approaches leverage *hybrid deep learning architectures* (e.g., CNN-LSTM) and employ *traditional metrics* (MSE, R^2) for hydrological forecasting. **However, they fall short because** they inadequately prioritize extreme event signatures in loss functions, leading to critical underestimation of flood peaks, and struggle with lead times >12 hours due to temporal dependency decay. **Therefore, there is a need to address** these limitations in extreme-value capture and forecast horizon **by developing** a *temporally-aware architecture* with: Extreme-focused loss function: Reweighting penalty terms for peak errors, Cascaded attention mechanisms: Multi-head layers targeting antecedent moisture signals and Lead-time progressive training: Ladder technique with horizon-specific decoders.

13. S, V. ., S, G. ., M, D. ., & S, K. . (2024). Rainfall Based Flood Prediction in Kerala Using Machine Learning. *International Journal of Intelligent Systems and Applications in Engineering*, 12(16s), 141–144. Retrieved from <https://www.ijisae.org/index.php/IJISAE/article/view/4800>.

Existing approaches leverage *standard ML algorithms* (e.g., LSTM-RNN, Random Forest, SVM) with *historical rainfall data* to achieve baseline flood predictions. **However, they fall short because** they: Rely solely on rainfall (1901–2018) without real-time monitoring, terrain, or hydrological data that miss critical flood drivers like soil saturation and river dynamics, Use coarse

annual/monthly data that cannot capture flash flood triggers or local variability and Optimize for overall accuracy (AUC/ROC) but fail to specifically model rare high-impact events. **Therefore, there is a need to address these limitations** in data comprehensiveness and event-specific modeling by **developing** a *hybrid real-time framework* that: Fuses satellite radar, IoT sensors, and terrain data with rainfall inputs, Embeds extreme-value loss functions in LSTM architectures to boost peak-flood sensitivity and Deploys edge-computing nodes for district-level nowcasting in high-risk zones like Kochi and Alappuzha.

14. Mingrui Shi, Hongyuan Fang, Yangyang Xie, Huihua Du, Saiyan Liu, Jean Marie Ndayiragije, Nannan Liu, A flood forecasting method coupling the CE-QUAL-W2 and PINN models, Journal of Hydrology, 10.1016/j.jhydrol.2025.133699, 661, (133699), (2025), <https://doi.org/10.1029/2023WR036380>.

Existing approaches leverage *data-driven ML models* (e.g., ANNs) for flood prediction and employ *physical rainfall-runoff models* for interpretability. **However, they fall short because** (1) ML models require extensive historical data (>5 years, 15+ flood events) unavailable for 97% of Japanese rivers; (2) physical models demand costly high-resolution basin data that degrade over time; and (3) both approaches fail to leverage domain knowledge under data scarcity, causing unreliable forecasts in climate-emerging flood zones. **Therefore, there is a need to address** these limitations in data efficiency and knowledge integration by **developing domain-informed hybrid architectures** (INNs) that: Embed hydrological priors (tank model concepts, rainfall-runoff relationships) directly into NN structures via custom layers, Implement physics-based regularization through penalty terms in loss functions and Optimize for sparse-data resilience by co-training on simulated rainfall-flood scenarios.

15. A. Kazadi, J. Doss-Gollin, A. Sebastian, and A. Silva, “FloodGNN-GRU: a spatio-temporal graph neural network for flood prediction,” *Environmental Data Science*, vol. 3, p. e21, 2024, doi:10.1017/eds.2024.19, <https://doi.org/10.1017/eds.2024.19>.

Existing approaches leverage *physics-based simulators* (e.g., LISFLOOD-FP) for high-fidelity flood modeling and employ *ML models* (CNNs/RNNs) for efficient prediction. **However, they fall short because** (1) physics models incur prohibitive computational costs (e.g., 6h/Hurricane Harvey simulation) that prevent real-time warnings; (2) ML solutions operate in isolated spatial/temporal dimensions, failing to capture coupled spatio-temporal dynamics; and (3) velocity fields are reduced to scalars, violating fluid conservation laws. **Therefore, there is a need to address** these limitations in efficiency, integration, and physical consistency by **developing FloodGNN-GRU**—a graph-based architecture that: Unifies spatio-temporal modeling via GNN-GRU fusion, Preserves vector physics using Geometric Vector Perceptrons for velocity and Achieves 1000× acceleration vs. physics simulators while maintaining accuracy.

Identified Major Gaps from the Literature Review

(A) Data-Related Gaps

- Limited data availability in many regions, especially ungauged or data-scarce basins
- Over-reliance on single data modalities (e.g. only rainfall, or single-date optical images)
- Lack of multi-temporal / dynamic data utilization
- Poor integration of remote sensing (SAR, optical, DEM) with in-situ/meteorological data

(B) Model-Related Gaps

- Exclusive use of shallow ML or simple time series models ignoring deep learning’s capacity

- Underutilization of hybrid or physics-informed AI architectures that embed hydrological constraints
- Poor capture of spatio-temporal dependencies (e.g. most models treat spatial and temporal separately)
- Inadequate treatment of extreme events in training (loss functions not peak-sensitive)

(C) Feature Engineering and Fusion Gaps

- Ineffective feature selection and decomposition (e.g. poor IMF classification in VMD)
- Ignoring multi-sensor fusion (optical, SAR, DEM)
- Lack of dynamic, context-specific feature alignment

(D) Deployment and Usability Gaps

- Limited real-time forecasting capabilities
- Heavy computational requirements hindering operational use
- Absence of accessible, user-friendly tools (e.g. web apps or dashboards for stakeholders)
- Lack of explainability that translates into actionable insights

Problem Statement

Floods remain among the most **destructive natural hazards** globally, causing widespread **loss of life, economic disruption, and environmental damage**. Effective **flood risk mapping** and **early warning** are essential for **disaster preparedness** and **mitigation**. However, current **flood prediction systems** face **critical limitations** that undermine their **real-world applicability**, especially in **data-scarce** and **ungauged regions** where **vulnerability** is highest. Existing approaches either rely on **computationally intensive physics-based hydrological models** that require **extensive ground calibration** and cannot deliver **real-time forecasts**, or on purely **data-driven machine learning models** that lack **physical interpretability** and struggle to **generalize** across diverse **hydrological contexts**.

Moreover, many existing solutions fail to fully exploit **multi-modal, multi-temporal data sources**. **Single-date optical imagery, univariate rainfall records, or sparse historical flood events** are often used in isolation, limiting models' ability to capture the **dynamic progression** of flooding. There is limited **integration** of **satellite-based remote sensing** (e.g., **SAR for cloud-penetrating observations, DEM for terrain modeling**) with **meteorological and hydrological data**. Even studies that combine data types frequently treat **spatial and temporal features** separately, failing to model **complex spatio-temporal dependencies** essential for **accurate forecasting**.

Feature engineering and **model design** also suffer from significant gaps. Many approaches rely on **shallow ML algorithms** or **monolithic deep learning models** without adapting to the **unique characteristics** of different data sources (e.g., **low-frequency vs. high-frequency flood signals**). Few models **embed domain knowledge** such as **hydrological constraints** directly into their architectures, limiting **physical consistency** and **interpretability**. Furthermore, **explainability** is often treated as a **superficial post-hoc analysis** rather than an **integrated feature** that can generate **actionable insights** for **disaster planners** and **infrastructure managers**.

Proposed Approach

This research aims to address these critical gaps by developing a hybrid flood risk mapping system that strategically combines machine learning for historical tabular data with deep neural networks designed to process multi-temporal satellite imagery. The proposed solution will embed hydrological knowledge to enhance generalizability and interpretability, leverage multi-sensor data fusion for richer environmental context, and deliver an operational, user-friendly interface via a Streamlit web application. By doing so, it seeks to provide accessible, real-time, and accurate flood risk assessments that support effective decision-making even in data-limited and high-risk regions.